



Fig. 6: Blue Gene, a supercomputer based on mobile architecture (Image courtesy: <http://en.wikipedia.org>)

biological computing systems since the engineering type is restricted to doing only what these have been programmed for. Biological computing style could be made possible in a few years, such as robots that easily drive and walk in the real world. However, a computer that is capable of thinking may probably take a few decades.

Some application areas

Autonomic computing is expected to simplify the management of computing systems and open doors to applications such as seamless e-sourcing, grid computing and dynamic e-business.

E-sourcing is the ability to bring in IT as a utility, at the time it is required and in the needed quantity to complete the work.

Some other autonomic computing applications areas are memory error correction, server-load balancing, process allocation, monitoring power supply, automatic updating of software and drivers, automated system backup and recovery, and prefailure warning.

Autonomic cloud computing is related to empowering cloud infrastructures and platforms so that these can take their own decisions to incessantly achieve their assigned jobs. Cloud systems are required to consistently deliver their functionalities and facilities to users without

any form of human intervention, interpretation and instruction.

Grid computing is one area where autonomic computing equipped with self-managing capabilities can add a lot of value, and there are many related projects in process.

University of Pennsylvania, the USA, is making a potent grid that targets to bring advanced methods of breast cancer diagnosis and screening to patients at a low cost. The grid is a utility-like service delivered over the Internet, enabling hundreds of

hospitals to store mammograms in digital form. Analytical tools that aid doctors to diagnose individual cases and discover cancer clusters in the population are also available.

Then, there is North Carolina Biometrics Grid, which is available to thousands of researchers and educators to facilitate boosting the speed of genomic research that is likely to result in new medicines to fight diseases and grow more nutritious foods to satisfy global hunger.

Better access to higher computing power via grid computing integrated with the implementation of open standards will allow researchers to work together more easily on complex issues, which should benefit all mankind.

Weather forecasting and protein folding, where intricate medical calculations are needed, are application areas that need computers to work 24/7, continuously for a couple of years.

Progressively autonomic computers will provide tools to analyse these complex problems. Systems with mobile architecture, such as Blue Gene (Fig. 6), will allow the study of phenomena happening in split seconds at an atomic scale.

Autonomic computing will be able

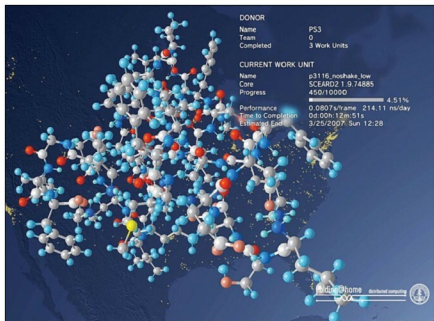


Fig. 7: Engineers are now beginning to understand that intuitive computing that calls for mimicking some or all aspects of the human brain (Image courtesy: <http://programming4.us/enterprise/24086.aspx>)